Inorganic Cumulative Exam
February 29, 2008

1. Permanganate ion (MnO_4^-) is strongly colored in solution yet Mn(II) and Mn(III) complexes are nearly colorless when dissolved in water in similar concentrations. Rationalize this observation in terms of the electronic structure of the metal centers.

2. $Co(NH_3)_5X^{2+}$ complexes show only weak absorbance bands in the visible but have strong shoulders in the near UV that shift to lower energy as X is changed from Cl to I, I having the lowest energy transition. What kind of electronic transition is responsible for the shoulder and how is the energy related to X?

3. Typically, an emission spectrum of a transition-metal complex consists of a single band in the 600-800 nm range skewed toward lower energy and 100-200 nm in width. $Cr(bpy)_3^{3+}$ and several other Cr^{3+} complexes have emission bands in the range 650-750 nm that are only 20 nm wide at half height. Rationalize this observation.

4. $Ru(bpy)_3^{2+}$, $Cr(bpy)_3^{3+}$ and related complexes have long-lived excited states that are either strongly oxidizing, strongly reducing or both. Rationalize this observation in terms of the electronic structures of the excited states.

5. A fundamental rule of photochemistry, sometimes called "Kasha's first rule of photochemistry" can be paraphrased as the following "Only colored molecules can undergo a photochemical reaction when exposed to visible light". What is the basis of this rule?